

Physics 111B: Optical Pumping

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(Dated: May 7, 2024)

The investigation delves into the quantum mechanical characteristics and magnetic behaviors of rubidium isotopes (Rb-85 and Rb-87) through the utilization of optical pumping and Zeeman effect measurements in a controlled laboratory setting. The experimental configuration incorporated Helmholtz coils to produce specific magnetic fields ranging from 0 to 4.85 Gauss, in conjunction with temperature-regulated conditions that impact the optical pumping procedure. The primary equipment consisted of a rubidium vapor cell, RF generator, and photodiode for the detection of resonance frequencies induced by varying magnetic fields and currents. Significant discoveries encompass the precise determination of the nuclear spins of Rb-85 and Rb-87, estimated to be around 2.575 and 1.553, respectively, closely approximating the theoretical values of 5/2 and 3/2. The ambient magnetic fields within the laboratory were gauged by the zero-current intercept of the calibration curve, yielding values close to 0.31 Gauss, in accordance with the anticipated influence of Earth's magnetic field at the geographical location of Berkeley, California. Furthermore, an intriguing observation of resonance in the absence of external RF fields indicated a quantum state degeneracy, offering valuable insights into state mixing and transition dynamics under zero-field circumstances. These outcomes furnish compelling evidence of the susceptibility of rubidium isotopes' quantum states to external magnetic fields and highlight the intricacies of manipulating atomic states. The experiment emphasizes the significance of precision in the experimental arrangement and alignment to accurately decipher atomic behaviors.

I. INTRODUCTION

Rubidium (Rb) belongs to the alkali metal group and exhibits typical characteristics of this group. One notable characteristic is the presence of a single valence electron, which plays a crucial role in determining the atomic and optical properties of the element. Naturally occurring rubidium consists of two isotopes, namely ^{85}Rb and ^{87}Rb . In the context of this particular experiment, our focus lies on the behavior of this lone electron at a principal quantum number of $n = 5$. Although the principal quantum number remains constant, the energy levels available to the electron are influenced by its angular momentum. In the field of atomic physics, the total angular momentum of an electron is a result of the vector sum of its orbital angular momentum ($\mathbf{L}$), spin angular momentum ($\mathbf{S}$), and, in the case of nuclei like rubidium, nuclear spin ($\mathbf{I}$).

The orbital angular momentum ($\mathbf{L}$) of the electron arises from its motion around the nucleus and is quantized in units of $\hbar$. On the other hand, the electron spin ($\mathbf{S}$) represents an intrinsic form of angular momentum associated with electrons and is also quantized in units of $\hbar$. The nuclear spin ($\mathbf{I}$) is a consequence of the combined angular momentum of protons and neutrons within the nucleus. In the case of rubidium isotopes, the nuclear spin values differ: ^{85}Rb has a nuclear spin of $I = 5/2$, while ^{87}Rb has a nuclear spin of $I = 3/2$.

Under idealized conditions where only Coulomb forces between the electron and the nucleus are considered, all states with the same principal quantum number n would be degenerate, meaning they would possess the same en-

ergy. However, in real atoms, this degeneracy is broken due to several small effects that are incorporated into the system's Hamiltonian. These effects include fine structure, which arises from spin-orbit coupling (an interaction between the electron's orbital angular momentum and its spin), and hyperfine structure, which is a result of the interaction between the electron's magnetic moment and the nuclear spin.

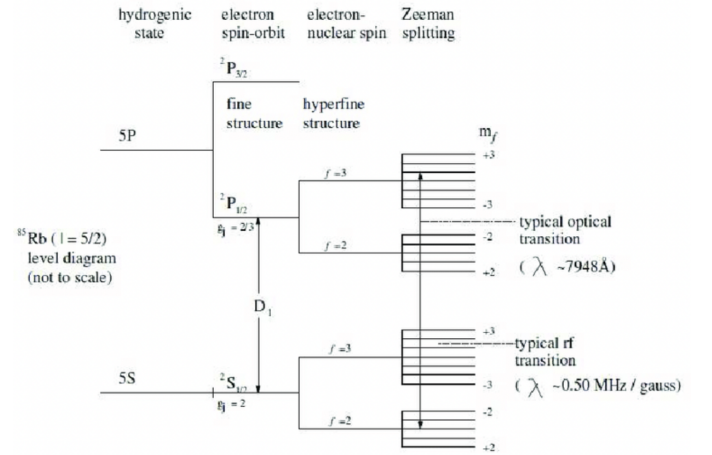


FIG. 1. Energy level diagram for ^{85}Rb showing the fine, hyperfine, and Zeeman splittings. [2]

This study focuses on analyzing the interactions and resulting energy splittings. The magnitude of these splittings can be significant; for example, the fine structure splitting is caused by spin-orbit coupling, dividing en-

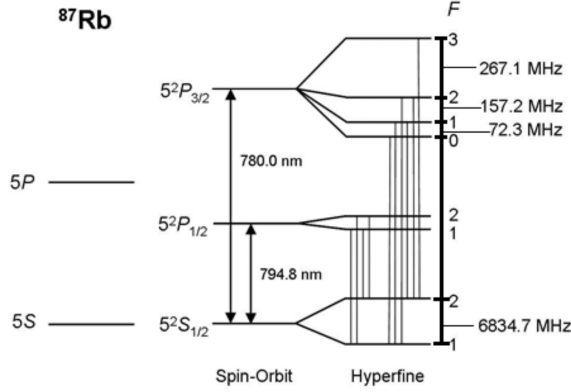


FIG. 2. Energy level diagram for ^{87}Rb indicating transitions and frequency shifts. [3]

ergy levels into sublevels based on different total angular momenta ($\mathbf{J} = \mathbf{L} + \mathbf{S}$). On the other hand, the hyperfine structure results from the interplay between the nuclear spin and the electron's total angular momentum ($\mathbf{F} = \mathbf{J} + \mathbf{I}$).

The energy splittings for the 5P excited states and the ground state 5S of rubidium are depicted in Figures 1 and 2. These figures showcase the transitions relevant to the research and demonstrate the division into various m_F sublevels under the influence of an external magnetic field, leading to Zeeman splitting.

A. The Electron-Coulomb Interaction

Rubidium, an alkali-metal element, possesses a lone valence electron that significantly impacts its chemical and physical characteristics. This electron engages in Coulomb interactions with the nucleus and other electrons, thereby determining its energy levels. Among these interactions, the electron-electron Coulomb repulsion stands out as particularly noteworthy. This repulsion is influenced by the orbital angular momentum, denoted as L , of the electron. Higher values of L indicate that electrons are located farther from the nucleus due to their increased angular momentum, resulting in a stronger centrifugal barrier. Consequently, this spatial arrangement enables outer electrons to shield inner electrons more effectively, thereby affecting the atom's ionization energy and reactivity. Our investigation concentrates on the $n = 5$ shell, exploring both the S state (where $l = 0$) and the P state (where $l = 1$) [5].

B. Fine Structure

Fine structure energy splitting in atoms emerges from relativistic corrections that link the electron's orbital angular momentum, denoted as $\vec{L}$, with its spin, denoted

as $\vec{S}$. This interaction introduces an additional term to the Hamiltonian, expressed as:

$$H_f \propto \vec{L} \cdot \vec{S} \quad (1)$$

where H_f represents the Hamiltonian associated with the fine structure interaction. This term considers the relativistic effects of the moving electron, which become significant at high electron velocities.

When examining the total angular momentum $\vec{J}$ of the electron, which is the vector sum of $\vec{L}$ and $\vec{S}$, we observe that the fine structure Hamiltonian commutes with $\vec{J}$:

$$[\vec{L} \cdot \vec{S}, \vec{J}] = [\vec{L} \cdot \vec{S}, \vec{L} + \vec{S}] = 0 \quad (2)$$

This is because when commuted with $\vec{L}$ you get $i\hbar(\vec{L} \times \vec{S})$ and when commuted with $\vec{S}$ you get $-i\hbar(\vec{L} \times \vec{S})$, cancelling out and indicating that $\vec{J}$ serves as a good quantum number, allowing states to be characterized by definite J values, resulting in energy level quantization where each level corresponds to a specific J . The fine structure splitting can be quantified using the following expression derived from perturbation theory:

$$\vec{L} \cdot \vec{S} = \frac{1}{2} (J^2 - L^2 - S^2) \quad (3)$$

By incorporating this into the energy calculations, the energy levels for states affected by fine structure are adjusted as follows:

$$E_{n,j} = -\frac{13.6 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] \quad (4)$$

where α denotes the fine structure constant, n stands for the principal quantum number, and j represents the total angular momentum quantum number resulting from the coupling of L and S .

In the case of the ground state of an electron in a hydrogen-like atom, such as the 5s state in rubidium, where $l = 0$ and $s = 1/2$, the value of j can only be $1/2$, resulting in a single energy level known as $2S_{1/2}$. On the other hand, the 5p state with $l = 1$ and $s = 1/2$ can have two possible values of j , specifically $1/2$ and $3/2$. These states are represented as $2P_{1/2}$ and $2P_{3/2}$, respectively.

The electron cloud surrounding the nucleus in S-states is spherically symmetrical, with zero orbital angular momentum. However, it is important to note that the electron probability density is not zero at the nucleus. This presence of electrons at the nucleus enhances the magnetic field generated by their motion, which in turn affects the nuclear magnetic moment through hyperfine interaction. On the other hand, P-states have one unit of orbital angular momentum and exhibit a nodal plane passing through the nucleus. In P-states, the electron probability density at the nucleus is zero. The spatial distribution of P-orbitals resembles a dumbbell shape, indicating that electrons spend less time in close proximity to the nucleus compared to electrons in S-orbitals.

The strength of the magnetic field created by electrons at the nucleus directly influences the hyperfine splitting. Since electrons in P-states are less likely to be found at the nucleus, the resulting magnetic field at the nucleus is weaker compared to S-states. As a result, the hyperfine interaction, which relies on this magnetic field, is weaker in P-states.

In S-states, where the orbital angular momentum (L) is zero, the total electron angular momentum is solely determined by the spin angular momentum ($J = S$). This simpler system leads to stronger hyperfine interactions due to the emphasis on spin-nuclear coupling. In P-states, the contribution of orbital angular momentum (L) to the total angular momentum (J) adds complexity to the vector sum of angular momentum, potentially resulting in partial cancellation or dilution effects in the magnetic interactions.

The transitions between these fine structure levels are characterized by specific wavelengths. For example, the transitions from $2S_{1/2}$ to $2P_{1/2}$ (referred to as the D1 line) and from $2S_{1/2}$ to $2P_{3/2}$ (known as the D2 line) correspond to wavelengths of approximately 795 nm and 780 nm, respectively. These transitions play a significant role in atomic spectroscopy and are crucial for the experimental investigation of atomic structure in optical pumping experiments.

C. Hyperfine Structure

The hyperfine splitting observed in atomic systems arises from the interplay between the magnetic moments of the nucleus and the electron. This interaction is described by the hyperfine Hamiltonian, which can be written as:

$$H_{\text{hfs}} = -\vec{\mu}_I \cdot \vec{B}_J - \vec{\mu}_J \cdot \vec{B}_{\text{ext}} \quad (5)$$

Here, $\vec{\mu}_I$ and $\vec{\mu}_J$ represent the magnetic moments of the nucleus and the electron, respectively. $\vec{B}_J$ denotes the magnetic field generated by the electron at the location of the nucleus, while $\vec{B}_{\text{ext}}$ represents an external magnetic field. [8]

The first term, $-\vec{\mu}_I \cdot \vec{B}_J$, corresponds to the intrinsic component of the hyperfine interaction and dominates in the absence of an external magnetic field. This term can be expressed as:

$$A\vec{I} \cdot \vec{J} \quad (6)$$

Here, A denotes the hyperfine coupling constant, while $\vec{I}$ and $\vec{J}$ are the operators for nuclear and total electronic angular momentum, respectively.

The second term involving $\vec{B}_{\text{ext}}$ accounts for the interaction of these magnetic moments with an external magnetic field. It is typically considered in experiments

that involve the manipulation or measurement of atomic states using external magnetic fields.

To determine the energy levels resulting from hyperfine splitting, one must take into account the total angular momentum $\vec{F} = \vec{I} + \vec{J}$, where $\vec{F}$ represents the total atomic angular momentum combining both nuclear and electronic contributions. The energy related to hyperfine splitting can then be computed using the formula:

$$E_{\text{hfs}} = \frac{A}{2} (F^2 - I^2 - J^2) \quad (7)$$

The energy shift for a state characterized by quantum numbers I , J , and F can be determined using the following expression:

$$H_{\text{hfs}}|J; I; F; m_F\rangle = A\hbar\frac{1}{2} (F^2 - I^2 - J^2) |J; I; F; m_F\rangle \quad (8)$$

$$= E|J; I; F; m_F\rangle \quad (9)$$

$$E|J; I; F; m_F\rangle = \left(\frac{A\hbar^2}{2}\right) [(F(F+1) - I(I+1) - J(J+1))] |J; I; F; m_F\rangle$$

This equation indicates that each combination of I , J , and F will result in a distinct energy shift, causing the previously degenerate levels to split.

When an external magnetic field is introduced, the hyperfine Hamiltonian must take into account the Zeeman effect, which modifies the Hamiltonian as follows:

$$H = H_{\text{hfs}} + \mu_B B_{\text{ext}} (g_J \vec{J}_z + g_I \vec{I}_z) \quad (10)$$

Here, μ_B represents the Bohr magneton, B_{ext} denotes the strength of the external magnetic field, and g_J and g_I are the g-factors for the electron and nucleus, respectively.

This additional term leads to further splitting and shifting of the energy levels, and we also notice that this should commute with m_F and so by diagonalizing in this subspace we can then solve for the Hamiltonian and get the exact energy perturbations. [4] In the case of a weak magnetic field, where the atomic structure is minimally affected, the energy levels can be approximated as:

$$E = g_F \mu_B B_{\text{ext}} m_F \quad (11)$$

Here, g_F represents the Landé g-factor for the state with total angular momentum F , and m_F corresponds to the magnetic quantum number associated with F . The value of g_F can be calculated using the formula:

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} \quad (12)$$

For a specific electronic state characterized by a given J , the hyperfine levels are established by the potential

values of F , which vary from $|I - J|$ to $|I + J|$. Each of these F -levels will possess its distinct energy due to hyperfine splitting, resulting in a complex spectrum of potential transitions within an atom. The atomic energy eigenstates for Rubidium isotopes, specifically ^{85}Rb and ^{87}Rb , are significantly impacted by the presence of hyperfine splitting within the theoretical framework. This influence stems from the nuclear spin (I) values associated with each isotope, where $I = \frac{5}{2}$ for ^{85}Rb and $I = \frac{3}{2}$ for ^{87}Rb . These nuclear spins play a crucial role in the hyperfine splitting observed in atomic spectra, particularly evident in the ground and first excited electronic states.

In Rubidium, the ground state corresponds to the $5s$ orbital (represented as $5^2S_{1/2}$), while the first excited state involves the electron transitioning to the $5p$ orbital (denoted as $5^2P_{1/2}$ and $5^2P_{3/2}$, reflecting different components of orbital angular momentum). The quantum mechanical analysis takes into account not only the electron's spin ($S = 1/2$) and orbital angular momentum ($L = 0$ for S-state and $L = 1$ for P-state) but also integrates the nuclear spin I . This integration results in a more intricate level structure due to the hyperfine interaction. For the $5s_{1/2}$ state where $J = \frac{1}{2}$, the possible total angular momentum quantum numbers F for the hyperfine structure can be $F = 2$ or $F = 3$. In the case of the $5p_{1/2}$ state where $J = \frac{1}{2}$, the allowed values for F are determined by the combination of J and the nuclear spin I , resulting in $F = 0, 1, 2, 3$. For the $5p_{3/2}$ state, with $J = \frac{3}{2}$, the possible values for F expand due to the greater J value, allowing for $F = 1, 2, 3, 4$. We can substitute these in to get the energy values, but to get the energy eigenstates, we can use these to represent in the m_F basis:

$$|\pm\rangle = |m_J = \pm\frac{1}{2}, m_I = m_F \pm \frac{1}{2}\rangle$$

From this we can then get neighboring zeeman state energy shifts, which should go as:

$$E = \frac{2.799 \times h \times B_{\text{ext}}}{2I + 1} \text{ MHz/G} \quad (13)$$

from which we can then get:

$$B_{\text{ext}} = \frac{\nu(2I + 1)}{2.799}$$

Hyperfine splitting is extremely important as in these types of states there is no exact F value, and thus for B_{ext} approaching 0, we can approximate F , and this should in turn make splitting dependent on m_F . Using our prior equations, we can now determine the energy eigenstate and energies for each isotope at every energy level for the hyperfine splitting in the absence of an external magnetic field. The table presented below displays the calculated values for each energy level along with their corresponding eigenstates. In the case of the ground state

energy, the values for A are approximately 1011.9 MHz and 3.4173 GHz for Rubidium-85 and Rubidium-87 respectively. Consequently, this results in an energy splitting of $4.18488214 \times 10^{-6}$ eV for both isotopes in the ground state energy level.

D. B Field

In our experimental setup, we utilize two coaxial coils with a radius a and a separation distance d . These coils are connected in series, ensuring that the same current i flows through both of them. To establish symmetry, we define the origin $z = 0$ at the midpoint between the two coils.

The magnetic field $B(z)$ along the axis of the coils can be determined using the Biot-Savart Law. This law states that the infinitesimal magnetic field $d\vec{B}$ at a point is given by the equation:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \vec{r}}{r^2} \quad (14)$$

Here, μ_0 represents the permeability of free space, I is the current, $d\vec{\ell}$ is the infinitesimal segment of the coil carrying the current, and $\vec{r}$ is the position vector from the coil segment to the field point.

Now, the Helmholtz coil is being considered. According to the Biot-Savart Law, it is known that the magnetic field $B(x)$ is given by the equation

$$B(x) = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}} \quad (15)$$

where μ_0 represents the permeability constant, I is the current passing through the coil, a stands for the radius, and x denotes the distance of the coil to a specific point. In this scenario, x can be expressed as either $\frac{d}{2} + z$ or $\frac{d}{2} - z$. By substituting I with NI to consider the number of turns in the coils, the combined magnetic field from the two coils is given by,

$$B_z(x) = \frac{\mu_0 N I a^2}{2} \left[\left(a^2 + \left| \frac{d}{2} + z \right|^2 \right)^{-3/2} + \left(a^2 + \left| \frac{d}{2} - z \right|^2 \right)^{-3/2} \right] \quad (16)$$

Moving forward, the second derivative of the magnetic field at $z = 0$ is calculated as

$$\frac{d^2 B}{dz^2} = \frac{768(d^2 - a^2)}{(4a^2 + d^2)^{7/2}} \quad (17)$$

When $d = a$, this derivative results in

$$\frac{d^2 B}{dz^2} = 0 \quad (18)$$

At $z = 0$ and $z = 2.5$, the values of B_z are $4.41 \times 10^{-4} T$ and $8.208 \times 10^{-7} T$ respectively. It is evident that even

under optimal circumstances, the magnetic field varies based on the location. Finally, the expression

$$B = \frac{0.9 \times 10^{-2} \times N \times I}{a} \quad (19)$$

is derived, providing the axial magnetic field produced by the Helmholtz coil.

E. Probabilities

In the realm of atomic and molecular physics, the distribution of probabilities among energy states holds significant importance in the prediction of system behavior under thermal equilibrium conditions. The allocation of each energy state's population adheres to the principles of statistical mechanics, particularly the classical Boltzmann distribution.

The Boltzmann distribution delineates the statistical spread of states for a system at thermal equilibrium with a consistent temperature. The likelihood $P(E_i)$ of a system occupying a state i with energy E_i is expressed as:

$$P(E_i) = \frac{e^{-\beta E_i}}{Z}$$

where $\beta = \frac{1}{k_B T}$, k_B represents the Boltzmann constant, T signifies the temperature, and Z denotes the partition function, which encompasses the summation across all feasible states of the system:

$$Z = \sum_i e^{-\beta E_i}$$

Within quantum systems, energy levels may exhibit degeneracy, implying that multiple states can share identical energy values. If an energy level E possesses a degeneracy g , the likelihood of the system occupying any of these g states is directly proportional to $g \cdot e^{-\beta E}$. Consequently, the probability of occupying a specific energy level E is influenced by its degeneracy.

$$\langle E \rangle = \sum_i E_i P(E_i) = \frac{1}{Z} \sum_i E_i e^{-\beta E_i}$$

Given that the hyperfine splitting in ^{87}Rb is about 6.835 GHz, the energy difference ΔE can be calculated using:

$$\Delta E = h \cdot 6.835 \times 10^9 \text{ Hz}$$

where $h = 6.626 \times 10^{-34} \text{ Js}$.

At room temperature (300 K), the partition function Z and the probabilities $P(E_i)$ are:

$$\beta = \frac{1}{1.381 \times 10^{-23} \cdot 300} \text{ J}^{-1}$$

$$Z = 1 + e^{-\beta \cdot \Delta E}$$

$$P(E_1) = \frac{1}{Z}, \quad P(E_2) = \frac{e^{-\beta \cdot \Delta E}}{Z}$$

At thermal equilibrium, it is highly likely to find atoms in lower energy states compared to higher energy states. Let's consider the case where the temperature is 40°C (313.15 K) and the transition occurs from the 5s state to the 5p state (with $S = 1/2$). For Rubidium-85, the transition energy is known to be 1.589 eV, while the value of k_B is $8.617 \times 10^{-5} \text{ eV/K}$. This information allows us to determine the ratio of the probabilities of finding atoms in the 5p orbital and 5s orbital, which is approximately 2.666×10^{-26} . Consequently, we can conclude that there will be a significantly larger number of atoms in the 5s state compared to the 5p state.

Now, let's examine the same ratio for Rubidium-85 in the context of transitions between hyperfine states in the electronic ground state. Specifically, we focus on the transitions between the states $F = 2$ and $F = 3$, where the transition frequency is given as 3.035 GHz. By converting this frequency to transition energy and keeping all other variables constant, we find that the ratio is approximately 0.9995, which is very close to unity. Furthermore, if we consider the ratio between neighboring Zeeman states in the electronic ground state, we observe a similar trend where the ratio is also close to unity. This suggests that in an atomic vapor cell at thermal equilibrium, there is almost no zero spin polarization.

Now, let's explore how we can generate spin polarization using optical pumping when the atoms are initially prepared in their ground state. By utilizing circularly polarized infrared light, we can drive transitions for the Rubidium atom across all $D1$ transitions. By selecting the z-axis (relative to the magnetic field) and employing left-handed polarization of the light, we can induce transitions that increase the magnetic quantum number m_F by one. When the atom undergoes spontaneous decay from its excited state, it emits an infrared photon and changes its magnetic quantum number by $-1, 0$, or 1 . Through multiple cycles of pumping, we observe that the final state of the atoms corresponds to the state with the greatest m_F value. In the event that the external magnetic field aligns with the light along the same axis, it results in the hyperfine state with the highest energy. Conversely, when the magnetic field is directed in the negative direction, it corresponds to the hyperfine state with the lowest energy.

F. What is Optical Pumping?

Optical pumping is a technique widely utilized to populate specific atomic states through the absorption and stimulated emission of light. In the case of Rubidium atoms, which possess remarkable hyperfine and Zeeman splitting characteristics, optical pumping is particularly advantageous for accurate magnetic field measurements and applications such as atomic clocks. [?]

In Rubidium atoms, optical pumping is employed to transfer the majority of atoms into a particular hyperfine and magnetic sublevel. For ^{87}Rb and ^{85}Rb , this typically involves the $|F = 2, m_F = 2\rangle$ and $|F = 3, m_F = 3\rangle$ states, respectively, under the influence of circularly polarized light that resonates with the D1 transition (from $5S_{1/2}$ to $5P_{1/2}$).

Initially, the atoms are distributed across various magnetic sublevels m_F within each hyperfine state F . The application of a magnetic field induces Zeeman splitting of these sublevels, which can be quantified by the equation:

$$\Delta E = g_F \mu_B B m_F \quad (20)$$

Here, g_F represents the Landé g-factor for the hyperfine state, μ_B denotes the Bohr magneton, B signifies the strength of the magnetic field, and m_F represents the magnetic quantum number.

Optical pumping, utilizing circularly polarized light (specifically σ^+ light), exclusively induces transitions where $\Delta m_F = +1$. This effectively "pumps" electrons to higher m_F states until they reach the maximum m_F value. Consequently, a significant population of atoms accumulates in the "stretched" state, which corresponds to the state with the highest m_F , as there are no higher m_F states available for transition. [7]

Once the stretched state is reached, further absorption of light is prohibited, resulting in a "dark state" where optical absorption is minimized. The atoms in this state can be probed using a radio-frequency (RF) magnetic field that is precisely tuned to induce transitions between different m_F levels within the same F state.

The frequency required to induce transitions between Zeeman sublevels is determined by the equation:

$$\nu = \frac{\Delta E}{h} = \frac{g_F \mu_B B}{h} \quad (21)$$

where h represents the Planck constant. By adjusting the external magnetic field B , it is possible to tune the RF resonance frequency, thereby gaining precise control over atomic state transitions. The total magnetic field experienced by the atoms is a combination of the applied magnetic field and any ambient magnetic field:

$$B_{\text{total}} = B_{\text{ext}} + B_{\text{amb}} \quad (22)$$

Here, B_{ext} denotes the field generated by the coils in the experimental setup, while B_{amb} is the magnetic field originating from the Earth's surroundings. The magnetic field produced by a coil can be computed using the Biot-Savart Law. For a Helmholtz coil configuration, the magnetic field at the center is expressed as:

$$B_z = 0.9 \times 10^{-2} \frac{Ni}{a} \text{ (T)} \quad (23)$$

In this equation, N represents the number of turns, i is the current flowing through the coil, and a is the radius of the coils.

The provided equations outline the methodology for calculating the resonance frequencies associated with transitions between Zeeman sublevels within a magnetic field. The resonance frequency is contingent upon the strength of the magnetic field, which encompasses both the external field generated by laboratory apparatus and the Earth's natural magnetic field.

The overall magnetic field (B_{total}) that impacts the atoms is the resultant vector sum of the external magnetic field (B_{ext}) produced by laboratory coils and the Earth's ambient magnetic field (B_{amb}):

$$B_{\text{total}} = B_{\text{ext}} + B_{\text{amb}}$$

The resonance frequency (ν) for transitions between Zeeman sublevels in the presence of a magnetic field is expressed as:

$$h\nu = \frac{2.799 \times h \times B_{\text{ext}}}{2I + 1} \text{ MHz}$$

Here, h represents Planck's constant, and I denotes the nuclear spin quantum number. The factor of 2.799 accommodates the gyromagnetic ratio specific to the system under consideration, standardized for convenient usage in MHz when B_{ext} is measured in Gauss.

By integrating the details of the coils and ambient magnetic field, the comprehensive frequency equation is formulated as:

$$\nu = \left(2.799 \times \left(\frac{0.9 \times 10^{-2} \times N \times i}{a} + B_{\text{amb}} \right) \right) \times \frac{m}{2I + 1}$$

In this equation (whose units are in MHz), N signifies the number of turns in the coil, i represents the current in amperes, a denotes the radius of the coils in meters, and m stands for the magnetic quantum number of the transition. B_{amb} corresponds to the ambient magnetic field in Gauss.

For an experimental arrangement with $N = 135$ turns and a coil radius $a = 27.5$ cm, these values are substituted to determine the specific frequency based on the provided current and ambient field values:

$$\nu = (12.3665 \times i + 2.799 \times B_{\text{amb}}) \text{ MHz} \div (2I + 1) \quad (24)$$

This equation enables the precise determination of the frequency required to induce magnetic resonance, taking into account the specific configuration and surrounding conditions. Then to get the zeeman transition frequencies, we can plug in known values to get $\nu_{\text{Rb}-85} = 2.29$ MHz and $\nu_{\text{Rb}-87} = 3.41$ MHz approximately. One thing to keep in mind is the Earth's magnetic field near Berkeley is about 0.48 Gauss which is the ambient magnetic field.

The Zeeman effect is highly sensitive to the overall magnetic field that the atomic or molecular samples are subjected to. Any misalignment results in a combination of the ambient and Helmholtz fields, which in turn

changes the effective field strength at the sample location. Given that the Zeeman splitting is directly proportional to the magnetic field strength, represented as $\Delta E = g\mu_B B$, any misalignment can lead to fluctuations in the measured splitting as the current in the Helmholtz coils varies. Consequently, this leads to less predictable resonance frequencies, making the interpretation of experimental data more complex. [9]

Achieving ideal alignment ensures a linear correlation between the Helmholtz coil current i and the total magnetic field strength. On the other hand, misalignment introduces magnetic field components that do not scale linearly with i , resulting in a non-linear relationship between the coil current and the observed Zeeman resonance frequencies.

Even if the Helmholtz coil is precisely adjusted to cancel out the ambient field along the optical axis, any misalignment can result in residual perpendicular components. These components represent the vector disparities between the ambient field and the precisely nullified component.

Consequently, the residual field is likely to be the Earth's field component that is perpendicular to the axis of the Helmholtz coil setup. The existence of a residual perpendicular magnetic field can lead to the broadening and asymmetry of spectral lines. This broadening occurs because different parts of the sample experience slightly varied total magnetic field strengths.

For accurate measurements, particularly in fields like quantum mechanics or atomic clocks, the presence of an unaccounted residual magnetic field can introduce systematic errors. These errors have the potential to distort the calibration of magnetic field sensors or impact the precision of g-factor measurements.

II. EXPERIMENTAL SETUP

A. Atomic Vapor Cell and Light

The experimental setup consists of a spherical glass bulb with a diameter of a few inches, internally coated with a thin layer of wax to prevent rubidium adhesion. Within the bulb, a low-density mixture of neutral buffer gas and a small amount of rubidium vapor is enclosed. The rubidium density in this controlled environment is adjusted by regulating the bulb's temperature using a resistive heater element. This heater is managed by a Heater Control Unit (HCU), which activates or deactivates the heating element based on feedback from a temperature sensor. This configuration minimizes temperature fluctuations, ensuring stable experimental conditions. It is important to highlight that although the heater is crucial for temperature regulation, it can produce a stray magnetic field that may affect Optically Detected Magnetic Resonance (ODMR) measurements. Therefore, during precise magnetic field measurements, the heater is turned off to prevent interference.

For optical pumping, light emitted from a lamp containing rubidium vapor, excited by an electronic discharge, is utilized. This light is specifically tuned to the D1 line of rubidium (with a wavelength of approximately 795 nm) to ensure efficient absorption by the rubidium vapor. Initially, the light passes through a polarizing filter to generate circularly polarized light, which is then filtered through a narrow-bandpass filter isolating the desired 795 nm wavelength. The filtered light is directed through the atomic vapor cell, and its absorption is monitored by a photodiode. The signal from the photodiode is amplified using an SRS SR560 amplifier, enabling the sensitive detection of light absorption variations, which are then displayed on an oscilloscope. This setup establishes an optical axis (z-axis) along which the light propagates, facilitating atomic excitation through optical pumping.

As for the magnetic resonance, after numerous rounds of optical pumping, the atoms eventually enter a state known as the "pumped" state, in which they cease to absorb the infrared light utilized during the optical pumping process. This state, referred to as the "dark" state, arises because all the atoms, having been pumped, possess the maximum m_F value. Since the atoms are already in their highest m_F value state, they are incapable of absorbing additional light that would elevate their magnetic quantum number. However, if we were to transition the light driving $D1$ transitions to $D2$ transitions, the pumped state would no longer be a dark state. This is due to the fact that when the optical pump induces $D2$ transitions, the atoms will transition from their excited state to either the ground state energy or their first excited state. Consequently, not all atoms in the pumped state will be in the state with the highest m_F value, and thus they can still absorb light.

By leveraging the concept of pumped states as dark states, we can identify radio-frequency magnetic transitions within the atom vapor cell. In a pumped state, the atoms do not absorb any light, enabling the photodetector to detect the maximum amount of light. Upon application of a resonant radio-frequency magnetic field, the atoms exit their pumped state, allowing them to absorb some light. As a result, the photodetector registers a decrease in the amount of light detected.

B. Enclosure and Setup

Examining Figure 3, we see that positioned adjacent to the vapor cell are two electromagnet coils aligned coaxially and connected in series. These coils are powered by an SRS DS345 function generator, inducing a sinusoidal current to generate a perpendicular RF magnetic field in relation to both the optical and external axial magnetic fields. This setup is crucial for facilitating Zeeman transitions within the rubidium vapor, which are essential for achieving the objectives of the experiment.

To ensure the reliability of the experiments, the va-

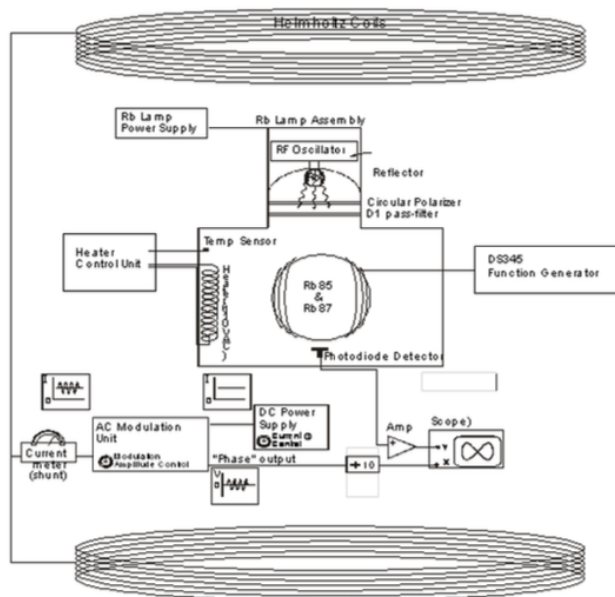


FIG. 3. The enclosure setup for optical pumping. [1]

por cell, along with its associated heating and optical components, is enclosed within a thermally insulated and light-proof container. This enclosure serves to minimize external light disturbances and uphold a stable internal temperature, which is imperative for obtaining precise and reproducible outcomes.

The external magnetic fields, pivotal for manipulating the Zeeman splitting of rubidium energy levels, are produced by two large Helmholtz coils positioned outside the insulated container. These coils generate a uniform magnetic field along the z-axis, which can be adjusted as per the experimental needs.

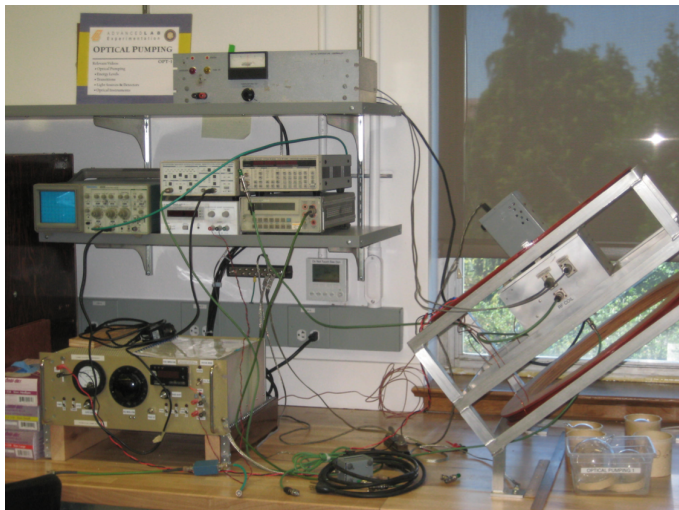


FIG. 4. A picture of the whole setup pulled from the lab manual. [1]

Interactions with the experimental setup as shown in Figure 4 primarily involve three control mechanisms:

1. Regulating the temperature of the heater to control the density of rubidium vapor.
2. Adjusting the current flowing through the Helmholtz coils to modify the strength of the external magnetic field.
3. Changing the frequency of the RF field to induce transitions out of the dark state in rubidium, enabling detailed investigations of atomic transitions and energy state distributions.

These controls are how we will examine the interplay between optical and magnetic fields within the rubidium vapor.

III. PROCEDURES

The initial step in the experimental calibration involves the establishment of a stable magnetic field using direct current. Specifically, a 1A current is passed through the Helmholtz coils to create a static magnetic field that determines the Zeeman energy differences. In order to manipulate the rubidium vapor inside the bulb, it is heated to 50°C using a resistive heater, which increases the vapor pressure of rubidium. Once the desired temperature is reached in the bulb, the heater is turned off to eliminate any magnetic interference it may cause. Subsequently, the RF frequency is swept across a 3 MHz range to identify the resonance frequencies of the rubidium isotopes, confirming the proper operation of the setup. As the bulb cools down, the variation in photodiode signal intensity is continuously measured, correlating with temperature changes from 50°C to 30°C, to comprehend the thermal dynamics affecting the experiment.

To improve the detection of Zeeman resonances, a 60Hz AC modulation is applied to the Helmholtz coils while maintaining a constant RF signal. This method helps in distinguishing the resonance peaks more effectively by converting the DC noise component into an AC signal, making it easier to detect signal variations caused by resonance absorption. Additionally, this setup allows for phase-sensitive detection of the signal, leading to a clearer identification of the resonance conditions as a Lissajous figure is observed on the oscilloscope which can be seen in Figure 5, forming a perfect figure eight at resonance alignment. When we are at lock-in frequency we expect the scope to look exactly like this and if we are slightly off resonance it should be a misshapen version of an infinity symbol, so basically the closer you are to resonance and the lock-in point the more it should look like a vertically symmetric infinity symbol.

The depth of modulation of the magnetic field is influenced by altering the magnitude of the 60 Hz modulation. Increasing the modulation depth can improve the visibility of the resonance peak by amplifying the

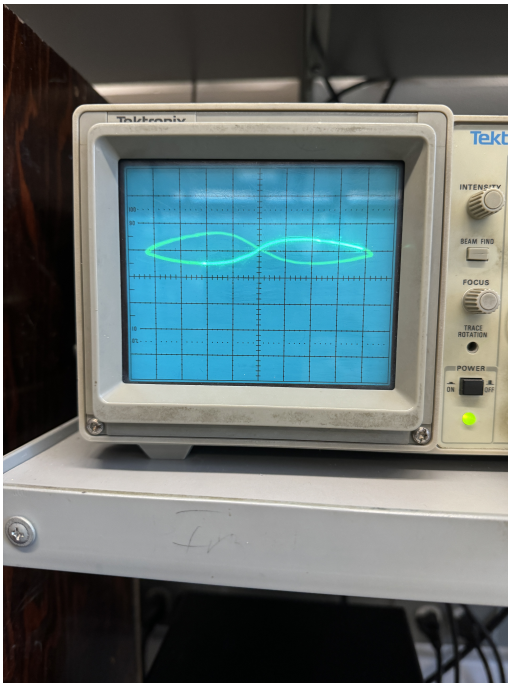


FIG. 5. The observed Lissajous figure.

signal amplitude at resonance. Nevertheless, excessively high modulation can widen the resonance peak, making it more challenging to accurately determine the center. The absorption characteristics are directly impacted by the density of rubidium atoms in the vapor cell. A higher density increases the likelihood of photon absorption, resulting in stronger resonance effects and a more pronounced signal. However, excessively high densities can lead to self-absorption, where rubidium atoms re-absorb emitted photons, as well as collisional broadening, both of which can obscure the clarity of the resonance peaks. Lastly, the power of the rf field interacting with the rubidium atoms is influenced by the amplitude of the rf signal. Higher amplitudes enhance the population of atoms transitioning to excited states, thereby strengthening the resonance signal. However, similar to the modulation depth, excessively high amplitudes can cause power broadening, ultimately reducing the resolution of resonance detection.

The measurements primarily focus on capturing the resonance frequencies at different levels of DC current through the coils, ranging from 0 to 2A, in both forward and reverse directions. This strategy not only aids in mapping out the resonance landscape but also acts as a safeguard against any external magnetic fields by zeroing the current. Furthermore, the measurement process is refined by deactivating the RF modulation and directly scanning the current to isolate the impact of ambient magnetic fields and detect any remaining resonance signals.

In order to investigate the dynamics of optical pumping, we utilize amplitude-modulated RF signals in a 1A

setting on the coils. By employing a 10Hz square wave modulation, we alternate between periods of RF activation and RF silence. This enables us to observe the temporal characteristics of the rubidium vapor's light absorption. Through this approach, we are able to determine the "pumping time" required for all atoms to reach the dark state, as well as the "relaxation time" needed for them to return to equilibrium once the RF field is deactivated. This methodology provides valuable insights into the efficiency and speed of optical pumping under different conditions.

The presence of a heater in the experiment can lead to the resonance frequencies exhibiting erratic behavior or instability. This phenomenon is likely attributed to the expansion and contraction of the components within the enclosure due to thermal effects, as well as variations in the physical properties of the gas as its temperature fluctuates. These thermal fluctuations have the potential to impact the magnetic field inside the enclosure by altering the magnetic susceptibility of the materials or causing slight displacements in the apparatus, which in turn affect the distribution of the magnetic field. By deactivating the heater, it is possible to stabilize the temperature and consequently mitigate the thermal fluctuations that give rise to the erratic resonance frequencies. However, it is important to note that turning off the heater may result in temperature drift, leading to changes in the density of the Rubidium. Additionally, lower temperatures can potentially decrease the rate of atomic collisions that contribute to the relaxation processes, thereby influencing the amplitude and clarity of the signal. When manipulating the current, it becomes evident that the resonance frequency tends to vary, with higher currents causing an increase and vice versa. Typically, two resonances are observed, and it is generally expected that the higher frequency resonance corresponds to 87 Rb due to the well-known frequency ratio of 0.6666. By adjusting the frequency, these resonances can be identified.

IV. RESULTS

The data presented in Figure 6 depicts the correlation between temperature and voltage measurements for two isotopes of Rubidium, namely Rb 85 and Rb 87, over a range of temperatures spanning from approximately 32.5°C to 50.0°C . Each data point on the graph represents the voltage needed to maintain optical pumping of the Rubidium vapor at a specific temperature.

In the case of Rb 85 (represented by blue dots), the voltage generally increases as the temperature rises. This suggests that a higher amount of energy is required to achieve optimal pumping as the temperature increases. On the other hand, Rb 87 (represented by green dots) exhibits a consistent voltage requirement across the entire temperature spectrum, albeit generally at lower voltages compared to Rb 85 at corresponding temperatures.

This discrepancy in voltage behavior between the two

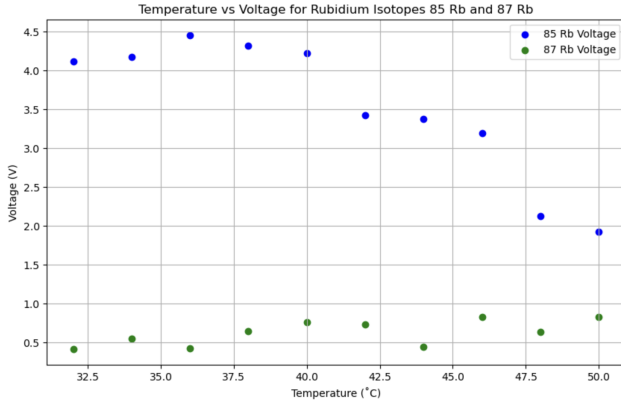


FIG. 6. The data for the variation in voltage for the 2 isotopes as we vary temperature.

isotopes indicates that they possess different energy dynamics. This difference could potentially be attributed to their distinct atomic structures and how they respond to changes in temperature within the experimental setup.

The resonant frequency obtained from the RF function generator reflects the absolute value of the energy difference (ΔE) between Zeeman levels, regardless of the field's orientation. Consequently, when creating plots, it is important to include both positive and negative currents on the frequency axis as seen in 7. The slope of each plotted line as seen in Figure 8 will indicate how the resonant frequency changes with the current passing through the coils, serving as a measure of the field generation efficiency per unit current. The intercept, which ideally should be zero in the absence of other magnetic influences such as the Earth's magnetic field, can reveal the presence of residual or misaligned magnetic fields. Since magnetic fields are vectors, any misalignment between the Helmholtz field and the ambient field will lead to a vector sum that deviates slightly from the expected field direction, potentially affecting the observed frequencies. Analyzing the residuals of the linear fit as seen in Figure 9 is crucial for detecting systematic errors or misalignments. Substantial residuals at zero current may indicate a static offset in the magnetic field, likely caused by ambient contributions that are not fully canceled out by the coils. Additionally, employing a weighted fit can help mitigate discrepancies and errors, particularly when dealing with small changes in the magnetic field.

The provided figures illustrate the frequency response of Rubidium isotopes 85 and 87 in relation to the current passed through Helmholtz coils. These graphs play a crucial role in comprehending the magnetic properties and the Zeeman effect exhibited by these isotopes under varying magnetic fields.

The initial plot (Figure 7) is a scatter plot that presents the individual measurements alongside their corresponding error bars. These error bars hold significant importance as they allow for the analysis of measurement precision and data reliability. Such plots are indispensable for

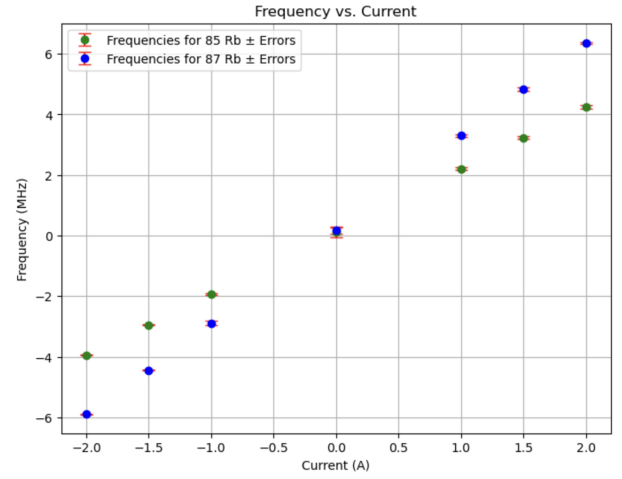


FIG. 7. The data for the frequency vs current.

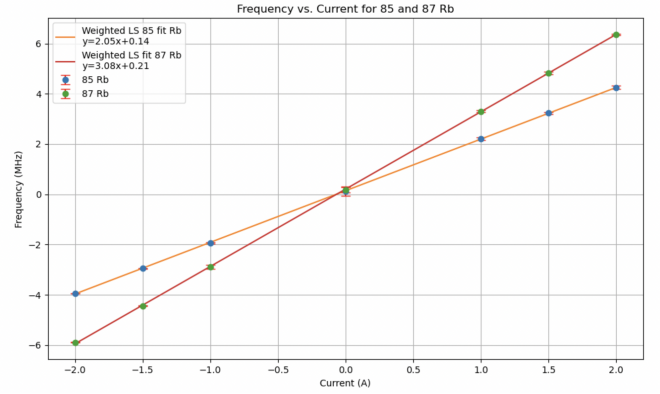


FIG. 8. The fitted line on the frequency vs current data.

evaluating the accuracy of the fit and for identifying any systematic errors or anomalies within the experimental setup.

On the other hand, the second plot (Figure 8) showcases the linear fits for each isotope and polarity, with the gradients indicating the sensitivity of the magnetic moment to the applied current. This arrangement aids in indirectly determining the gyromagnetic ratio and provides valuable insights into the magnetic characteristics of the isotopes under investigation.

In this investigation, it is deemed appropriate to examine the relationship between frequency and current by means of a linear fit, considering the expected direct correlation between the magnetic field generated by Helmholtz coils and the current flowing through them. This linear correlation is believed to mirror the influence of the Zeeman effect on atomic energy levels, which is influenced by the external field.

The adequacy of this model fit can be assessed using the chi-squared (χ^2) statistic, which quantifies the sum of the squared differences between the observed and model-predicted values, normalized by the variance of

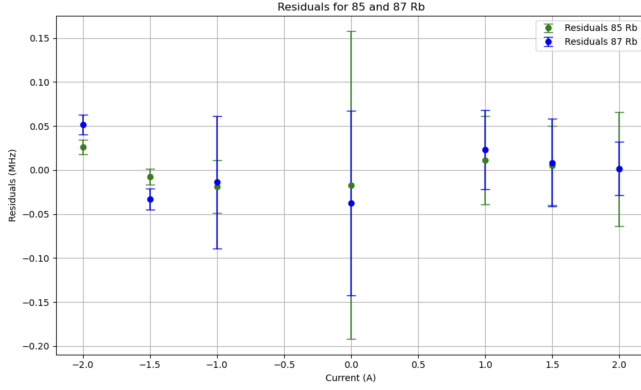


FIG. 9. The residuals of the plotted line versus the actual point for the frequency vs current graph.

the observed values. A low χ^2 value in relation to the degrees of freedom indicates a good fit, signifying that the model effectively represents the data. In this particular scenario, the chi-squared test offers a statistical measure along with a p-value, where the p-value signifies the likelihood that any observed disparities between the model and the data are random. For Rubidium 85, the chi-squared statistic was 11.9997 with a p-value of 0.0348, indicating a reasonably good fit despite potential significant deviations. Conversely, for Rubidium 87, the higher chi-squared value of 30.0444 and a notably low p-value suggest a less satisfactory fit, hinting at potential systematic errors or deficiencies in the linear model for this isotope.

By plotting residuals, which represent the variations between observed frequencies and those predicted by the linear model, a visual and quantitative approach can be employed to further evaluate the quality of the fit. The presence of large residuals or discernible patterns in the residual plot may indicate inadequacies in the model or systematic errors within the data. Given the minimal errors associated with data points that could render these disparities imperceptible on the primary plot, focusing on residuals enables a more comprehensive error analysis and model assessment. Here we do not see any clear trend with the residuals, so there is no discernible pattern.

To improve data analysis, merging data from both polarities of the current could lead to a more reliable dataset by utilizing symmetry and increasing the effective sample size. This approach may aid in stabilizing the fit and minimizing the impact of outliers or noise in individual measurements.

When it comes to handling errors in data fitting, incorporating the derived error estimates as weights in a weighted least squares regression can result in a fit that takes into account the varying confidence levels associated with each measurement. On the other hand, assuming equal weights for all data points and then evaluating errors based on the fit residuals might uncover potential underestimations in the reported measurement errors. The discrepancies observed in the fit results and

error estimates between these two methodologies could be attributed to non-uniform error distribution or other underlying factors that are not captured by a basic linear model. Although, the tendency seems to be that the fit results agree relatively well when accounting for error, because the order of the error is very similar across data points, and the lack of a huge discrepancy in error will contribute to a fit that is very similar when including errors as weights. Of course in general they won't exactly agree, because accounting for weight errors allows a more precise line to be made, which is how we have chosen to plot our lines here, but generally speaking the already present strong linear trend tends to dominate and there is not much variation from that regardless of weights.

Our data, in conjunction with the lines of best fit, can be utilized to determine the nuclear spin of both isotopes in two distinct manners. Initially, we analyze the ratios of all the frequencies. The frequency ν can be theoretically expressed as:

$$\nu = \frac{12.3665 \cdot i + 2.799 \cdot B_{\text{amb}}}{2I + 1} \text{ MHz}$$

Here, i represents the current flowing through the Helmholtz coils, B_{amb} denotes the ambient magnetic field, and I signifies the nuclear spin quantum number. By dividing the frequencies of the two different isotopes at the same current setting, we obtain:

$$\frac{\nu_{85}}{\nu_{87}} = \frac{2I_{87} + 1}{2I_{85} + 1}$$

Taking the average of this ratio across all measured frequencies yields:

$$\frac{2I_{87} + 1}{2I_{85} + 1} \approx 0.6612$$

Remarkably, this result closely approximates the expected theoretical value of $\frac{3/2}{5/2} = 0.666$. Further analysis involves directly determining the nuclear spin for each isotope by examining the slope of each line of best fit. The relationship derived from the equation for ν is:

$$m = \frac{12.635}{2I + 1}$$

Solving for I gives:

$$I = \frac{\left(\frac{12.635}{m}\right) - 1}{2}$$

Applying this to the weighted slopes of the best-fit lines for both isotopes, we find that $I_{85} \approx 2.575$ and $I_{87} \approx 1.552$, values which closely approach the expected nuclear spins of 5/2 and 3/2 for ^{85}Rb and ^{87}Rb , respectively. From these results, we can see that the ratio and

half-integral is more than sufficient to get a very strong estimate on nuclear spins. Considering that systematic errors may occur due to imperfections in the alignment of the Helmholtz coil with respect to the Earth's magnetic field, assumptions made regarding the uniformity of the magnetic field, or variations in the calibration of measurement instruments, these factors have the potential to deviate the results from the theoretical expectations. Now we can get the errors.

The nuclear spin I can be expressed in terms of the slope m of the linear fit as:

$$I = \frac{\left(\frac{12.635}{m}\right) - 1}{2}$$

In order to determine the error in I resulting from the error in m , the standard error propagation formula is utilized, where the error σ_I is computed as:

$$\sigma_I = \sqrt{\left(\frac{\partial I}{\partial m}\right)^2 \sigma_m^2}$$

The derivative of the function $I(m)$ with respect to m is:

$$\frac{\partial I}{\partial m} = \frac{\partial}{\partial m} \left(\frac{\frac{12.635}{m} - 1}{2} \right) = -\frac{12.635}{2m^2}$$

Hence, the error in I , denoted as σ_I , due to the error in m , σ_m , is given by:

$$\sigma_I = \sqrt{\left(-\frac{12.635}{2m^2}\right)^2 \sigma_m^2}$$

This leads to the final expression for the error in I :

$$\sigma_I = \frac{12.635}{2m^2} \sigma_m$$

This formula enables the determination of the uncertainty in the nuclear spin I based on the measured uncertainty σ_m in the slope of the fitted line.

The discrepancies between the errors in the nuclear spin I for ^{85}Rb and ^{87}Rb and their respective calculated values are notable. The error in I for ^{85}Rb is found to be 0.00698, while for ^{87}Rb it is 0.00605. In comparison, the calculated values for I are 2.57533 for ^{85}Rb and 1.55286 for ^{87}Rb .

Ideally, these values should fall within the error bounds of the theoretical nuclear spins of 2.5 for ^{85}Rb and 1.5 for ^{87}Rb . However, the discrepancies suggest that there are factors contributing to the observed differences. Possible sources of error include unaccounted systematic errors in the experimental setup, such as misalignment in magnetic fields, calibration inaccuracies in measurement instruments, or flawed assumptions in the model used for data analysis. While the statistical errors from the fitting process are minimal, they may not fully encompass the variability present in the data, particularly if the linear

model's underlying assumptions are violated. Additionally, the theoretical model employed to determine I may not be entirely precise or may necessitate further adjustments that were overlooked in the initial evaluation.

The magnetic fields obtained from the measurements of both isotopes of Rubidium (^{85}Rb and ^{87}Rb) at currents of ± 1 A are presented above, along with their corresponding uncertainties. The magnetic field for ^{85}Rb at +1 A is found to be 4.849486 ± 0.001318 G, while for ^{85}Rb at -1 A, it is 4.245407 ± 0.010987 G. Similarly, the magnetic field for ^{87}Rb at +1 A is 4.849074 ± 0.001760 G, and for ^{87}Rb at -1 A, it is 4.233290 ± 0.007334 G.

The calculated magnetic field based on the dimensions of the coil is 4.418182 ± 0.091679 G. A comparison of the measured magnetic fields with the theoretical calculation reveals that there is consistency within the isotopes at both positive and negative currents. However, there is a noticeable deviation between the measured fields and the coil-based calculation, particularly for positive currents.

The higher accuracy, indicated by the lower uncertainty, in the measurements at positive currents suggests that the setup is more stable or better calibrated under these conditions. The discrepancy between the experimental measurements and the coil-based calculations could be attributed to various factors. It is possible that the actual distribution of the magnetic field does not align perfectly with the theoretical model, especially if the coil deviates from the assumed ideal shape or size. Additionally, other magnetic fields present in the environment or misalignments in the setup may be influencing the measurements. Furthermore, errors in the calibration of the equipment used to measure current or the magnetic field could also contribute to the observed discrepancies.

The magnetic field B produced by a Helmholtz coil can be calculated using the given formula:

$$B = \frac{0.9 \times 10^{-2} \times N \times i}{a}$$

where N represents the number of turns, i represents the current, and a represents the radius of the coil.

To determine the error in B , denoted as ΔB , we can utilize the following formula, taking into account the errors Δi in the current and Δa in the radius:

$$\Delta B = B \sqrt{\left(\frac{\Delta i}{i}\right)^2 + \left(\frac{\Delta a}{a}\right)^2}$$

In the specific case where $N = 135$, $a = 27.5$ cm = 0.275 m, $i = 1.00$ A, $\Delta i = 0.01$ A, and $\Delta a = 0.005$ m (half of the smallest digit), we can substitute these values into the formula to calculate ΔB :

$$\Delta B = \frac{0.9 \times 10^{-2} \times 135 \times 1.00}{0.275} \sqrt{\left(\frac{0.01}{1.00}\right)^2 + \left(\frac{0.005}{0.275}\right)^2}$$

The Bohr magneton, denoted as μ_B , is a crucial physical constant that characterizes the magnetic moment of

an electron due to its orbital or spin angular momentum. It is expressed as:

$$\mu_B = \frac{e\hbar}{2m_e}$$

where e represents the elementary charge, $\hbar$ is the reduced Planck constant, and m_e stands for the electron mass. As per the Committee on Data for Science and Technology (CODATA), the current accepted value for the Bohr magneton is $\mu_B = 9.2740100783 \times 10^{-24}$ J/T with a standard uncertainty of $0.0000000028 \times 10^{-24}$ J/T. This corresponds to a relative standard uncertainty of roughly 3×10^{-10} , or approximately 0.3 parts per billion.

Due to the remarkable precision of the known value of μ_B , the uncertainty associated with it is notably smaller compared to the uncertainties typically encountered in experimental measurements within a standard physics laboratory environment. For instance, in our specific experiment concerning the determination of the magnetic field generated by Helmholtz coils, errors are more likely to arise from measurements of physical dimensions (such as the coils' radius and the current passing through them) rather than from the fundamental constants utilized in the computations. Consequently, the uncertainty in μ_B is deemed insignificant and does not exert a substantial impact on the overall uncertainty of our experimental outcomes.

When examining the energy levels of atoms in magnetic fields, disregarding the Zeeman energy may result in notable inaccuracies, especially in precise measurements or in the presence of strong magnetic fields. The Zeeman effect causes the energy levels of atomic states to split into various sublevels, contingent upon the magnetic quantum number m_F . The energy alteration due to the Zeeman effect can be expressed as:

$$\Delta E = g_F \mu_B B m_F$$

where g_F represents the Landé g-factor for the hyperfine state, μ_B denotes the Bohr magneton, B stands for the magnetic field intensity, and m_F signifies the magnetic quantum number. This equation illustrates that the change in energy is directly proportional to the magnetic field and the magnetic quantum number.

In scenarios where the magnetic field is relatively feeble, the energy shift induced by the Zeeman effect may be insignificant compared to the hyperfine splitting and other energy magnitudes in the scenario. Nevertheless, with an escalation in the magnetic field intensity, or for states featuring higher magnetic quantum numbers, this approximation becomes less accurate. The error stemming from neglecting the Zeeman energy can be approximated as the ratio of the Zeeman energy shift to the pertinent energy scale, such as the hyperfine splitting:

$$\text{Error} \approx \frac{\Delta E}{\Delta E_{\text{hyperfine}}}$$

For standard laboratory magnetic fields and low- m_F states, this discrepancy might be minor, yet it amplifies

with both the field intensity and m_F . Consequently, in meticulous measurements or in experiments involving potent magnetic fields or elevated m_F values, disregarding the Zeeman energy can result in substantial disparities between theoretical forecasts and experimental findings. But generally, from earlier, we know the order of bohr magneton to be so small relative to our values that it would likely create an unnoticeable discrepancy relative to other uncertainty measurements.

Resonance was unexpectedly observed even when there was no applied radio-frequency (RF) field present. This intriguing phenomenon occurred specifically when a reverse current of 0.08 Amps was applied, effectively nullifying the external ambient magnetic field, B_{ext} . In this unique state, the rubidium atoms were unaffected by any external magnetic influences.

The occurrence of resonance under these zero-field conditions can be attributed to the quantum mechanical properties of the atomic states. Without an external field to differentiate the energy levels of the magnetic quantum states (m_F), these states converged to a single energy level. Ordinarily, each m_F state would occupy a distinct energy level, influenced by B_{ext} . However, in the absence of such a field, these energy levels merged.

This degeneracy allowed for the mixing of states over time, facilitated by thermal and quantum interactions among the atoms. Consequently, electrons were able to transition out of the "dark state," which is a state that does not absorb light at the operational frequency of the experiment. As a result, resonance was observed even without the presence of an external RF field, as electrons transitioned between mixed states that temporarily became energetically favorable.

When examining the observed pumping and decay times, these times are affected by a range of factors, which complicates precise calculations. To make accurate estimations of these times, one must take into account the frequency and likelihood of resonant transitions induced by the RF field, the statistical distribution of decay times from the excited state, and the frequency of these decays.

The dynamics of transitions under the resonant RF field play a crucial role. These transitions rely on the interaction between the characteristics of the RF field and the quantum mechanical properties of the atoms. Additionally, the decay process from the D1 transition in rubidium follows a statistical distribution, necessitating thorough modeling for precise prediction of decay times. The rate of decay is influenced by various factors, such as the energy levels involved and the external conditions of the experiment.

Therefore, although we can observe and measure these times, accurately predicting them requires a comprehensive understanding of atomic transitions and their interaction with the external RF field, as well as a statistical analysis of decay processes.

Utilizing the y-intercept of the linear fit equation allows us to estimate the ambient magnetic field in the laboratory. By setting the current through the Helmholtz

coils to zero, the frequency attributed solely to the ambient magnetic field can be determined using the formula:

$$\nu = \frac{2.799 \cdot B_{\text{amb}}}{2I + 1} \text{ MHz}$$

To solve for the ambient magnetic field, the equation can be rewritten as:

$$B_{\text{amb}} = \frac{\nu(2I + 1)}{2.799}$$

By substituting the y-intercept frequency values obtained from the regression analysis for both isotopes at $i = 0$, the ambient magnetic field can be calculated. The results are as follows:

- For Rb 87: $B_{\text{amb}} = 0.310$ Gauss
- For Rb 85: $B_{\text{amb}} = 0.301$ Gauss

These values offer an estimation of the earth's magnetic field, which is essential for comprehending the external factors affecting the observed Zeeman effect. From the values we can see that they are relatively consistent with each other, and propagating our errors in I into this, we see that for Rb 85 we have an error of 0.00197 and for Rb 87 we have an error of .00248, and comparing this to the range of earth's values we see that it falls within those values which is a strong estimate, but at Berkeley the ambient field should be around 0.5, and this value is not within error bounds as it is too small. Precision and accuracy of methods primarily depends on minimization of error sources and here we could say (as mentioned earlier) there are many possible sources for statistical and systematic error to leak in, and thus there can be better experiments to more accurately determine the value. Precision has to do with the ability to do multiple trials and get the same value repeatedly which would mean that if other students were getting similar values then this would be a precise experiment.

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In the conducted experiment, the pumping time was determined to be 400 ± 50 ms, while the relaxation time was found to be 70 ± 30 ms. This was accomplished by adjusting each time block on the oscilloscope to 50 ms and monitoring the signal transitions. The correlation between the pumping and relaxation times is anticipated, given that the pumping process, which entails population inversion, typically requires more time compared to the spontaneous relaxation of Rubidium atoms. To make a precise comparison between these empirically obtained values and theoretical estimations, it would be necessary to possess comprehensive data regarding the Rubidium density in the bulb and the intensity of the pumping light.

V. CONCLUSION

The investigation effectively employed optical pumping methods to investigate the quantum mechanical properties of rubidium isotopes. Important results of the research included the precise determination of nuclear spins and the evaluation of the Earth's ambient magnetic field at Berkeley, in good agreement with theoretical predictions for the nuclear spin, but an undervalued estimate for the ambient field. These discrepancies were likely a result of experimental limitations such as equipment misalignment and calibration errors as has been explained in the results section. A notable discovery was the identification of resonance even in the absence of an external RF field, which can be attributed to the degeneracy of quantum states and their thermal mixing. These results underscore the complexities of atomic interactions and the

impact of external influences on quantum states, shining light on the importance of meticulous experimental

setup and methodology to effectively analyze quantum mechanical phenomena in atoms.

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